Anant Khanna

itsanantk@gmail.com | itsanantk.github.io | linkedin.com/in/itsanantk | github.com/itsanantk

Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science in ***Computer Engineering*** *- Dean's List 2025* — *Sept. 2024 – May 2028*

Experience

Machine Learning Engineer | *Agriprix* — May 2026 – Present
- Own the **ML and data stack** end-to-end at an early-stage B2B agricultural marketplace
- Built automated **price-intelligence pipelines** in Python that ingest, deduplicate, and normalize multi-source listing and transaction data into per-commodity price benchmarks
- Developing **time-series price forecasting** models with seasonality, regional, and quality-grade features
- Designed and shipped **REST APIs**, **PostgreSQL** schemas, and **React** features for listings, orders, and supplier onboarding

Machine Learning Software Engineer Team Lead | *UBC UAS* — Sept. 2024 – Present
- Integrated **YOLOv11 (PyTorch)** pipeline in Python and C++ for real-time **autonomous object detection** on embedded systems travelling at 50+ km/h, achieving **98% detection accuracy** and **+30% mAP**
- Engineered a **metric learning pipeline** for target re-identification, computing mean vectors from Vision Transformer **(ViT)** embeddings to enable cosine similarity matching invariant to lighting and orientation
- Implemented **TensorRT FP16 quantization** to compress model weights, reducing memory bandwidth usage by **50%** while maintaining inference precision within **0.5%** of the baseline

Lead Developer | *UBC Finds* — Oct. 2025 – Present
- Lead development of a real-time campus lost-and-found platform built on **React/Next.js** and **Supabase (PostgreSQL)**, shipping live map rendering and item status updates
- Architected **event-driven state synchronization** over Supabase realtime channels, eliminating race conditions between asynchronous user reports and live map state through atomic writes

Programming Instructor | *Code Ninjas* — May 2025 – Aug. 2025
- Designed and delivered **15+** project-based C#/JS curricula and led logic-building activities for **120+** students

Projects

Wealth Council | *Python, FastAPI, React, Google Gemini, Fetch.ai* — Mar. 2026
- Engineered an agentic investment platform, winning an award at the ProduHacks x Fetch.ai hackathon
- Architected a Multi-Agent Orchestration Layer using **Fetch.ai uAgents** to asynchronously compute portfolio risk metrics and integrated FinBERT **(NLP)** to perform sentiment analysis on live financial news streams
- Integrated Google Gemini to synthesize final reports, streamed to a React frontend in real time via **SSE**

GuardCam: AI Dashcam | *Python, React Native, WebSockets, OpenCV* — Mar. 2026
- Developed an edge-to-cloud driver monitoring system to detect micro-sleeps, utilizing a custom **TCP WebSocket bridge** to bypass HTTP overhead for low-latency video streaming
- Implemented **MediaPipe** and **OpenCV** on a headless Modal GPU server to map a 468-point 3D face-mesh, calculating Eye Aspect Ratio (EAR) to trigger sub-second haptic alerts

FPGA-Based Neural Network Inference Accelerator | *Verilog, FPGA, C++, ARM* — July 2025
- Designed and implemented a **single-layer neural network** (**MAC unit**) on a **DE1-SoC FPGA** using **Verilog**
- Developed a **memory-mapped C++ driver** on the ARM HPS for low-latency data streaming, offloading compute to the FPGA and enabling efficient hardware acceleration under **embedded Linux**
- Achieved **3,000** inferences/sec with a **300%** performance speedup over ARM-only execution by optimizing **HPS–FPGA** communication

Technical Skills

Hardware/RTL: FPGA Design (Intel Quartus, QuestaSim, Platform Designer), Verilog, SystemVerilog, ARM SoCs
Languages: Python, C, C++, C#, SQL, Java, JavaScript, RISC-V & ARM Assembly
Technologies: Linux, PyTorch, TensorFlow, TensorRT, OpenCV, YOLO, React, Next.js, Supabase, PostgreSQL, Git